EDA Mini Project - Exploratory Data Analysis on Electronic game

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Project Objective

The objective of this EDA project is to conduct a comprehensive analysis of the video game market to understand its core characteristics, identify key market winners, and determine the critical factors contributing to their success, ultimately aiming to generate actionable insights for future decision-making and pinpoint the most promising market segments for strategic entry.

Dataset Description

The dataset for this EDA project was sourced from Kaggle.com. It is a concise yet powerful dataset, comprising over 16,000 rows and 11 columns of historical video game data spanning from 1980 to 2016. It provides essential attributes such as platform, genre, publisher, and regional sales figures, which are crucial for analyzing market with variable identification recognized as below:

Nature	Type	Variable	Data Type
Qualitative	Descriptive	Name	object
	Categorical	Platform	object
		Genre	object
		Publisher	object
Quantitative	Discrete	Rank	Integer
		Year	Float
	Continuous	NA_Sales	Float
		EU_Sales	Float
		JP_Sales	Float
		Other_Sales	Float
		Global_Sales	Float

Data Cleaning

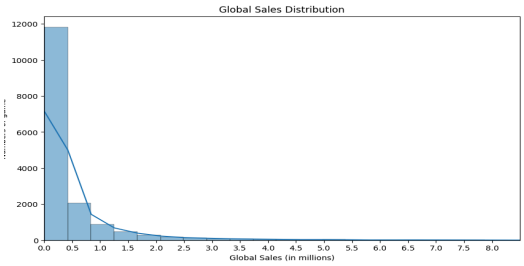
In the data cleaning phase of the project, we focused on identifying and handling missing or duplicated data points to ensure the quality and reliability of the dataset.

First, the dataset was checked for duplicate entries using `df.duplicated()`, and the total count of duplicates was recorded. Next, we specifically addressed missing values within the crucial 'Year' and 'Publisher' columns. To prioritize data integrity, games with missing 'Year' information were identified and sorted by `Global_Sales` to understand the potential impact of removing high-value entries. The top 10 games missing year data but having significant sales figures were specifically reviewed. For records missing the 'Publisher' information, those values were globally imputed with the string 'Unknown'. Finally, all rows with missing values in the 'Year' column were removed from the dataset using `dropna(subset=['Year'])`. As a final step in preparation for analysis, the data type of the 'Year' column was converted from a floating-point number (float) to an integer (int) for cleaner representation and accurate time-series analysis.

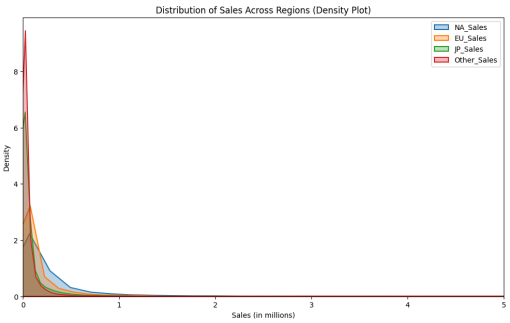
Analysis and Findings

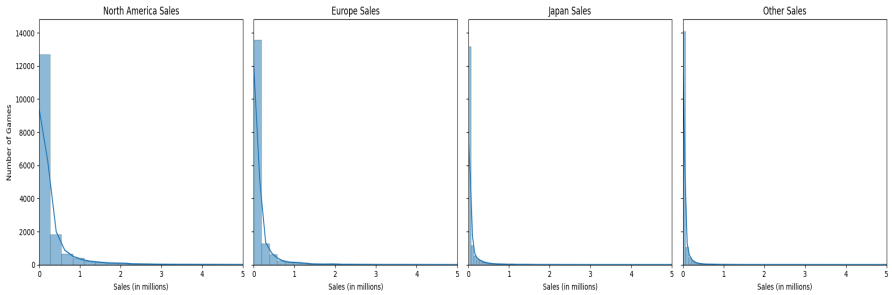
i. Univariate Analysis

-Global Sales Distribution



The analysis of the global sales distribution reveals a market characterized by extreme imbalance, effectively a "winner-takes-all" environment. As illustrated in the figure above, the distribution is highly right-skewed. While the mean sales might be around 0.5 million units, over 75% of all games sell less than this figure, highlighting that successful game are exceptions rather than the norm.

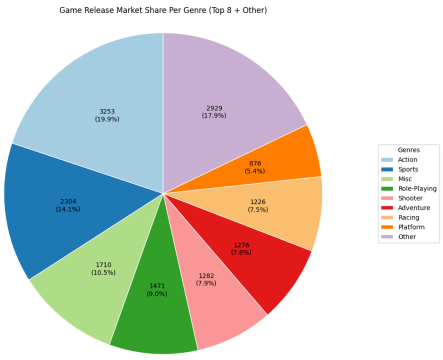




Comparing the sales distributions across the four primary regions (North America NA, Europe EU, Japan JP, and Other regions) highlights both shared market dynamics and unique regional characteristics.All four regions exhibit highly right-skewed distributions and every region's total sales volume is largely dependent on a small number of high-selling games.

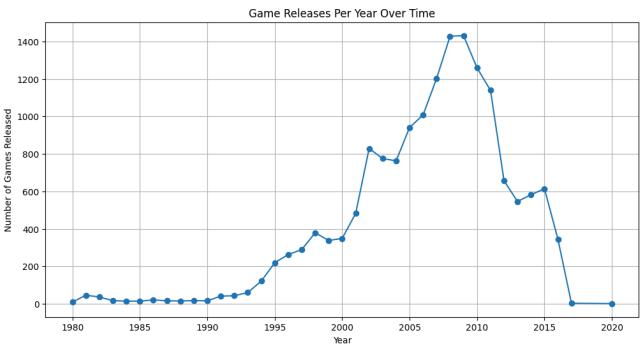
However, there are differences in the degree of volatility. The Japan market exhibits the lowest skewness, while other regions have the highest skewness.

-Market Share Per Genre



Based on the dataset, the top five genres by game count are Action, Sports, Misc (Miscellaneous), Role-Playing, and Shooter. When analyzing the top two genres, Action games (3,253 releases, 19.92%) significantly outnumber Sports games (2,304 releases, 14.11%), comprising approximately 41% more volume than the runner-up. This indicates that the Action genre holds a dominant position in the market share concerning the volume of releases.
Despite this prominence, the overall market distribution is more balanced than initially expected; no single genre holds an absolute monopoly. The market share is relatively fragmented, requiring five to six genres combined (Action, Sports, Misc, Role-Playing, Shooter, Adventure) to account for around 70% of total releases, which reflects a degree of market diversity within the gaming industry.
On the other end of the spectrum, we can see the lower-ranking game genres, such as puzzle, strategy, and fighting games. These genres each have release counts below 1,000, with market shares only accounting for approximately 3.5% to 5%. Notably, the genre with the lowest release count, puzzle games at 571, is only about one-sixth the volume of action games. This highlights the significant scale difference that exists between different genres in the market.

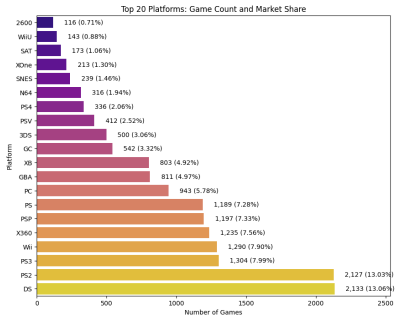
-Game releases per year over time from 1980 to 2020



The analysis of game release counts over time, spanning from 1980 to 2016, reveals that the market experienced its most prolific period of game releases between approximately 2007 and 2011. This surge corresponds directly to the middle of the seventh console generation, a time when multiple major platforms (PS2, PS3, Wii, Xbox 360, DS) were all simultaneously active and competing intensely in the market.

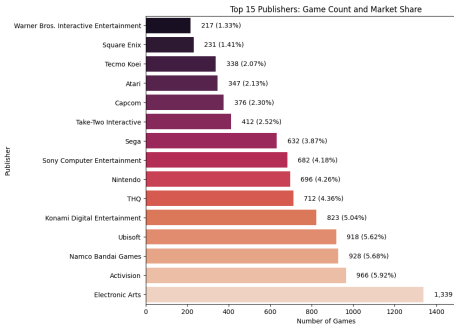
Following the 2011 peak, there is a marked and sustained decline in the number of annual releases. This shift suggests a move towards a "quality over quantity" strategy in the industry, possibly driven by higher development costs for next-generation consoles and the rise of mobile gaming.

-Top platform and Game Publishers



Our analysis reveals that home consoles were the primary drivers of the gaming market between roughly 2000 and 2013. Specifically, the overlapping competitive period from 2005 to 2013—involving major consoles like the PS2, PS3, Wii, and Xbox 360—created a historical peak in game releases.

Within this landscape, the Nintendo DS stands out as a unique case study. While technically a handheld console, it not only dominated the portable market but also became the single platform with the highest volume of game releases, recording 2,133 (just edging out the PS2 at 2,127). This was largely due to its low development costs and massive user base.

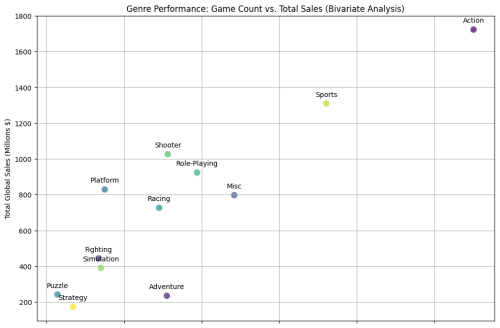


To understand who the market winners are, we analyzed the total number of game releases by each publisher. Our initial analysis suggests that the publishing market is dominated by a few major giants, such as Electronic Arts (EA), Activision, and Ubisoft. EA leads significantly with 1,339 games released (8.2% market share), while the top three publishers combined account for nearly 20% of the market share among the top 15.

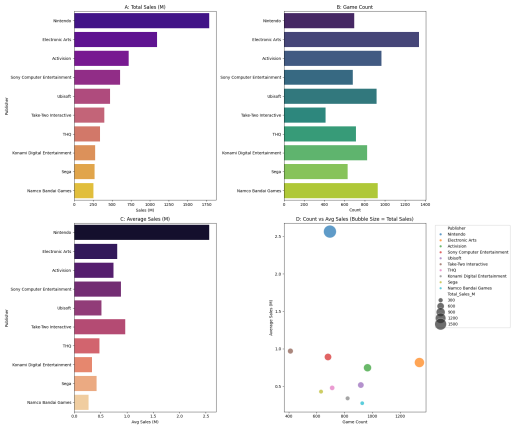
However, focusing solely on the volume of releases has limitations, as quantity alone does not measure true market success or profitability. Therefore, to identify the real market winners, our next step will be to explore which publisher's games achieved the highest average sales volumes.

ii. Bivariate Analysis

-Global sales and publisher

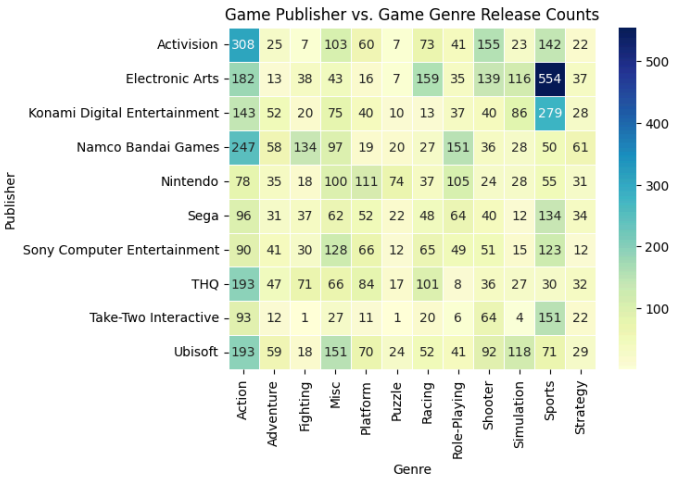


This report analyzes game genres by release count and total sales, showing that Action, Sports, and Shooter games sell the best, while Strategy, Adventure, and Puzzle sell the least. An interesting point is that more releases do not always mean more sales. For instance, Shooter games rank first in sales despite being fifth in releases, proving they are very popular. The Misc category has many releases but only average sales. Platform and Adventure games are also interesting; Platform games have fewer releases but good sales, suggesting that a few high-quality games can make a big difference in total revenue.



Next, we analyze different publishers' total sales, average sales, and game counts. Nintendo, EA, and Activision are market winners. Nintendo clearly leads in total sales, becoming the market leader. Even though Nintendo has fewer releases than EA and Activision, its high game quality gives it the best average sales. EA and Activision achieve good sales by releasing many games (volume strategy). Take-Two also has fewer releases but focuses on quality, ranking second in average sales. Finally, Namco Bandai Games releases many games but has the lowest average sales among the top ten, suggesting their games might not be popular with many players.

-Publisher Strategies and Genre Distribution

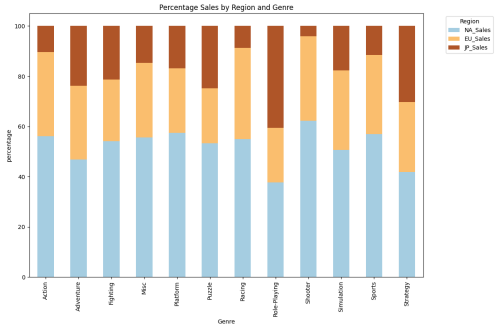


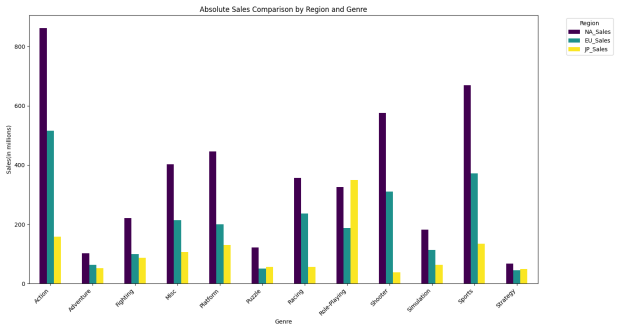
This heatmap analysis reveals clear differences in how companies approach game releases. EA focuses heavily on Sport games, dominating that market with 554 releases, nearly a third of the total for that genre. Most publishers concentrate on popular, high-selling types like Action, Shooter, Sport, and Role-Playing.

In contrast, market leader Nintendo cleverly avoids the top three highest-selling genres and instead focuses on Platform, Role-Playing, and Misc. The success of these top publishers is not accidental; they use precise market positioning and focus resources on areas of high demand or specific strengths like strong game IPs to secure their market positions.

iii. Multivariate Analysis

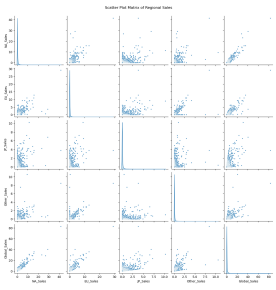
-Relationship between Sales ‘s region and Genre





This chart breaks down the percentage of sales for different game genres across three regions: North America (NA), Europe (EU), and Japan (JP), showing clear regional preferences. NA sales (light blue) are the largest share for most genres, especially for Shooter and Sports games, where they dominate the market. EU sales (orange) are consistently the second-largest portion across nearly all genres. JP sales (brown) are typically the smallest overall share, but they make up a significantly larger percentage for specific genres like Role-Playing and Platform games, often exceeding the EU share in those categories.

From the chart, it is evident that Shooter games are distinctly unpopular in the Japanese market (very small brown section), despite the genre's high global sales ranking. This highlights that the game types preferred in the Japanese market are significantly different from the global mainstream market. This cultural difference clearly impacts regional game sales and publisher strategies.



A multivariate analysis of cross-regional sales trends, visualized in the "Cross-Regional Sales Trends" chart, reveals three critical insights into global market dynamics:

1. Majority of Video Games Have Low Sales Volumes: The analysis shows that the vast majority of titles cluster near the origin point of the sales charts. This visually confirms that while a few "blockbuster" games achieve massive global sales, the typical video game release yields relatively low sales volumes across all regions.
2. North American and European Markets Share Very Similar Tastes: There is a strong positive correlation observed between the sales figures in North America (NA) and Europe (EU). Data points track very closely along a shared trend line, indicating that games that sell well in NA are highly likely to perform similarly well in EU, and vice versa. This suggests that publishers can often employ a unified strategy for these two Western markets.
3. Relative Independence of the Japanese Market: In contrast to the NA/EU correlation, sales trends for Japan (JP) show a much weaker or even independent relationship when plotted against Western sales figures. Games that are top sellers in NA and EU often have only moderate or low sales in Japan, while specific Japanese-preferred genres can perform exceptionally well there but fall flat in the West. This highlights the unique cultural preferences of the Japanese market, which operates with significant independence from the global mainstream.

Challenges and Limitations

During the Exploratory Data Analysis (EDA) project, several challenges and limitations were identified that potentially impact the scope and precision of the findings. One primary difficulty lay in data cleaning, as crucial 'Year' and 'Publisher' columns contained missing values (Null/NaN). Although we applied imputation and removal strategies, this data loss might introduce subtle biases or affect the sample size. Another major limitation is the dataset's timeframe, which only extends up to 2016. Given the significant

industry shifts since then (e.g., the rise of mobile gaming and subscription models), insights derived from this data may not fully represent current market dynamics. Furthermore, the data granularity is aggregated by year, lacking specific months or dates. This insufficient detail hinders a deeper analysis of short-term phenomena like launch-week hype or seasonal trends. Finally, relying solely on a single data source (Kaggle) introduces potential biases inherent in that specific compilation method, and the dataset did not include all gaming platforms, meaning the analysis of "market winners" might be skewed towards the traditional console and PC gaming sectors.

Future work and Limitations

Future work aims to move beyond EDA by developing a predictive model for global sales trends and merging external data sources to enrich the dataset. However, these efforts are currently constrained by limited project timelines, data granularity issues (e.g., lack of monthly data), and the fact that the project's primary scope was set for exploration, not production-ready model building.

Conclusion:

This EDA project successfully achieved its objectives, providing a clear perspective on the highly competitive video game market. By utilizing data-driven analytical methods, we have identified high-potential growth areas for the group and established a solid foundation for optimizing future game development and publishing decisions.

Reference

https://www.kaggle.com/datasets/volodymyrpivoshenko/video-game-sales-dataset

Work Distribution

Code, PPT, Report: Ian , Larry